

Antenna Datasheet

Product OC: YE0010AA
Version: 2.3
Date: 2026-05-18
Status: Released

Product Name: 4G Adhesive Mount Low Profile Dipole External Antenna
Key Features:
Frequency Band: 698–2700 MHz
Dimensions: 152 mm × 18 mm × 11.1 mm
Efficiency: Up to 79.5 %
RoHS Compliant
IP66

Overview

The Quectel YE0010AA is a high-performance, low-profile external antenna engineered for reliable 4G LTE connectivity, while also offering backward compatibility with 3G, 2G, and LPWA networks. Covering a wide frequency range of 698–2700 MHz, this dipole antenna ensures stable performance across all major global cellular bands.

The antenna features a versatile design, with a self-adhesive mounting pad that allows for easy and secure attachment to various device enclosures. It measures 152 mm × 18 mm × 11.1 mm and is lightweight. The YE0010AA offers excellent radiation efficiency (up to 79.5 %) and omnidirectional coverage, with minimal interference from the internal environment of the host device. Designed for harsh environments, this antenna features an IP66 rating, providing effective dust and water resistance, and operates within a temperature range of -40 °C to +85 °C. It comes standard with a 1-meter RG174 cable and an SMA male connector.

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: Free Space

1.1. Electrical

Electrical	
Frequency Range	698–2700 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

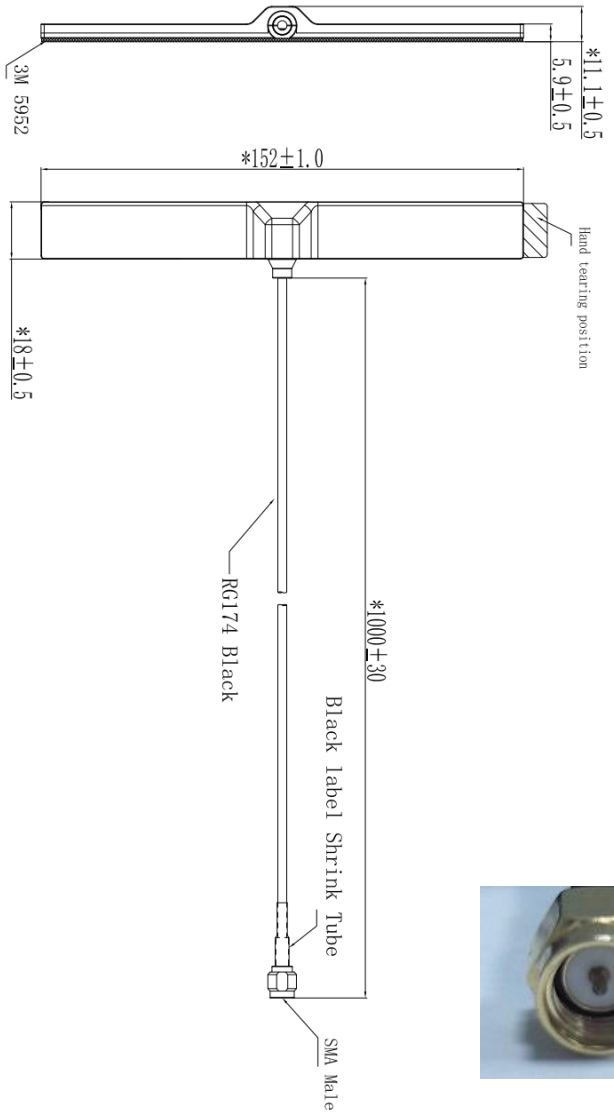
Electrical – Detail

Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR		3.9	2.1	2.6	-	2.5	2.2	1.8	1.6	-	-	-
Max. Return Loss(dB)		-4.5	-8.8	-7.0	-	-7.4	-8.3	-10.5	-12.5	-	-	-
AVG Eff. (%)		50.1	70.3	41.0	-	48.9	44.0	49.3	50.5	-	-	-
AVG.AVG Gain (dB)		-3.1	-1.5	-4.0	-	-3.1	-3.6	-3.1	-3.0	-	-	-
Max. Peak Gain (dBi)		2.2	2.0	1.5	-	2.4	1.5	2.1	2.7	-	-	-
VSWR		≤ 3.9										
Return Loss		≤ -4.5 dB										
Peak Gain		≤ 2.7 dBi										

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	152 mm × 18 mm × 11.1 mm
Material & Color	PC + ABS & Black
Cable Type & Color & Length	RG174 & Black & 1000 mm
Connector Type	SMA Male
Mounting Type	Adhesive
Weight	Typ. 29.8 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
Ingress Protection (IP) Rating	IP66
RoHS Compliant	Yes

2 Drawing

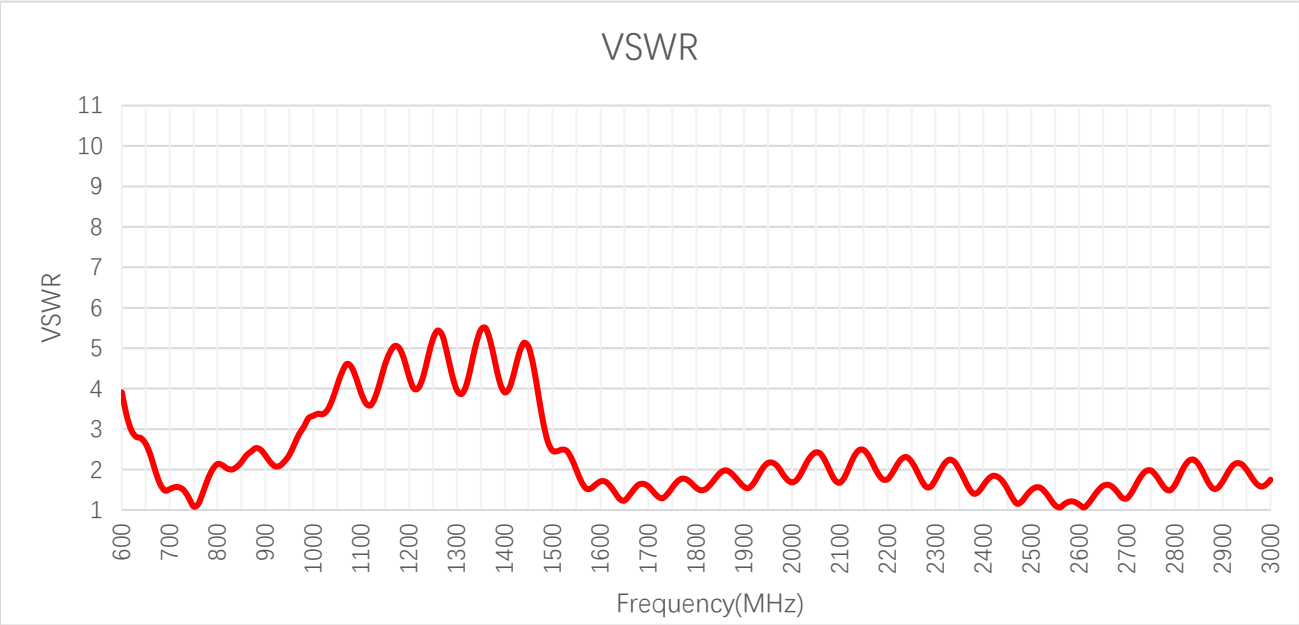

3M 5952
*11.1±0.5
5.9±0.5
*152±1.0
Hand tearing position
*18±0.5
RG174 Black
*1000±30
Black label Shrink Tube
SMA Male



3 Detailed Performance

3.1. S-Parameter Test

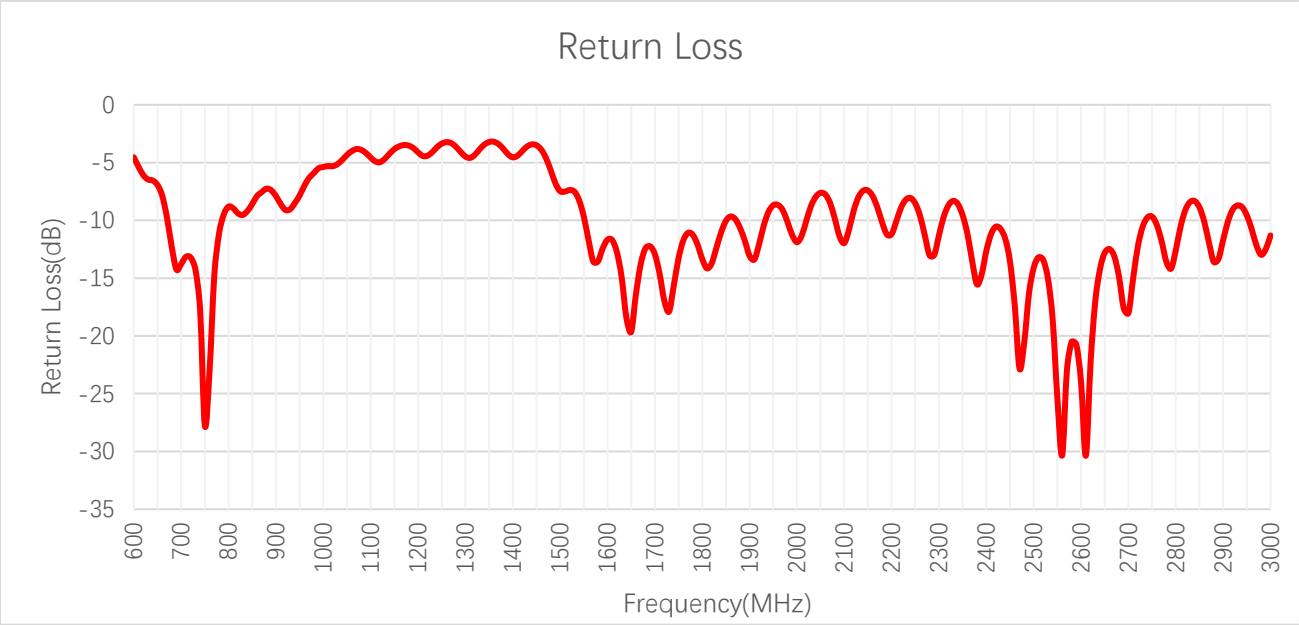
3.1.1. VSWR



VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
VSWR	3.9	2.8	1.6	2.0	2.4	2.6	-	1.5	1.4	1.8
Frequency (MHz)	1950	2140	2350	2450	2600	2690	4700	5000	5500	6000
VSWR	2.2	2.5	2.0	1.5	1.1	1.3	-	-	-	-

3.1.2. Return Loss

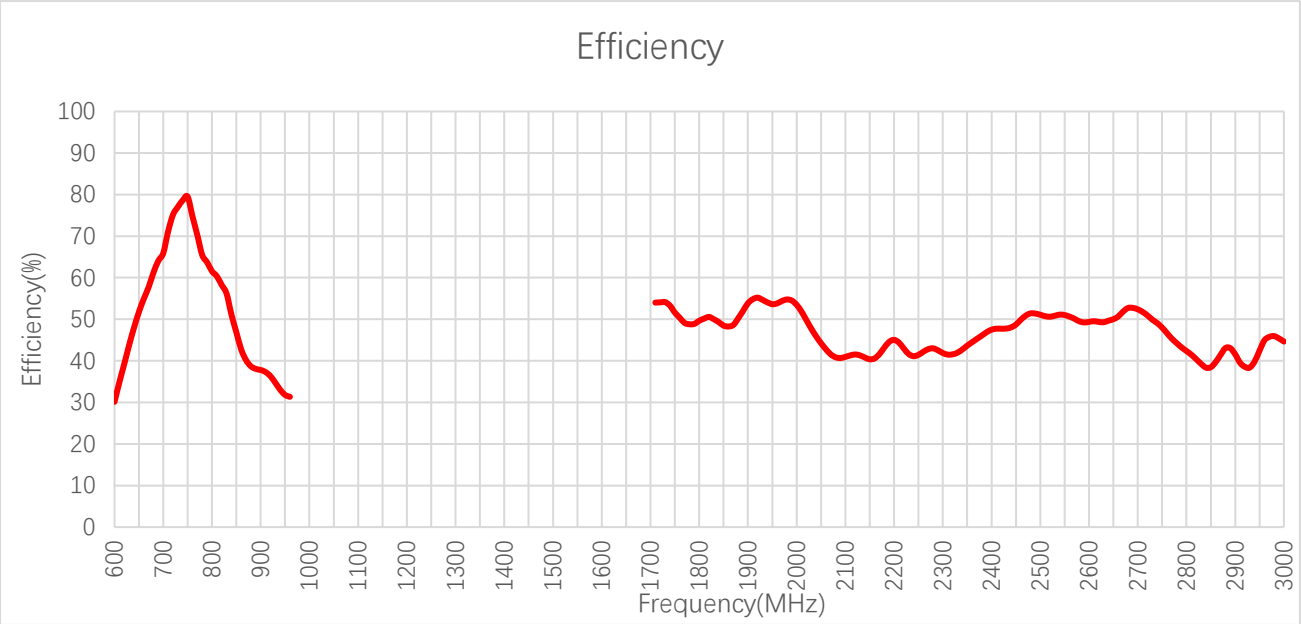


Return Loss (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Return Loss (dB)	-4.5	-6.5	-13.2	-9.5	-7.9	-7.0	-	-14.6	-15.7	-10.7
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Return Loss (dB)	-8.7	-7.4	-9.6	-13.6	-24.0	-17.6	-	-	-	-

3.2. Radiation Performance Test

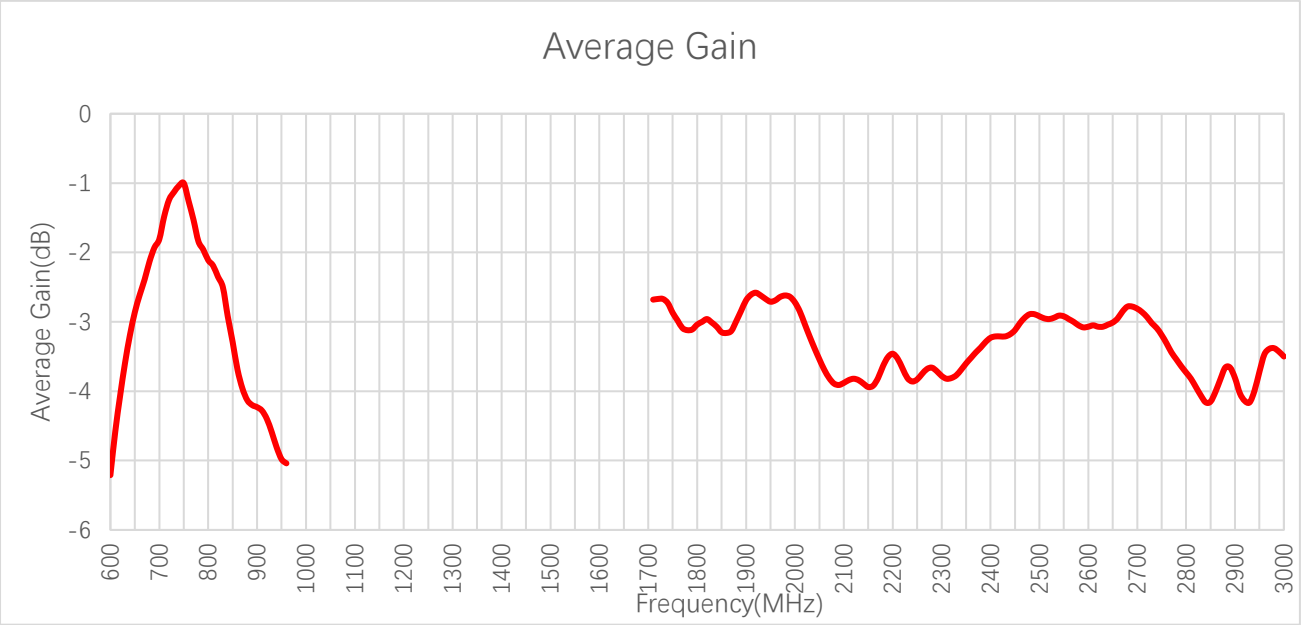
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Efficiency (%)	30.1	43.8	71.1	56.2	37.8	31.3	-	54.0	53.3	50.2
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Efficiency (%)	53.6	40.8	43.7	48.7	49.4	52.7	-	-	-	-

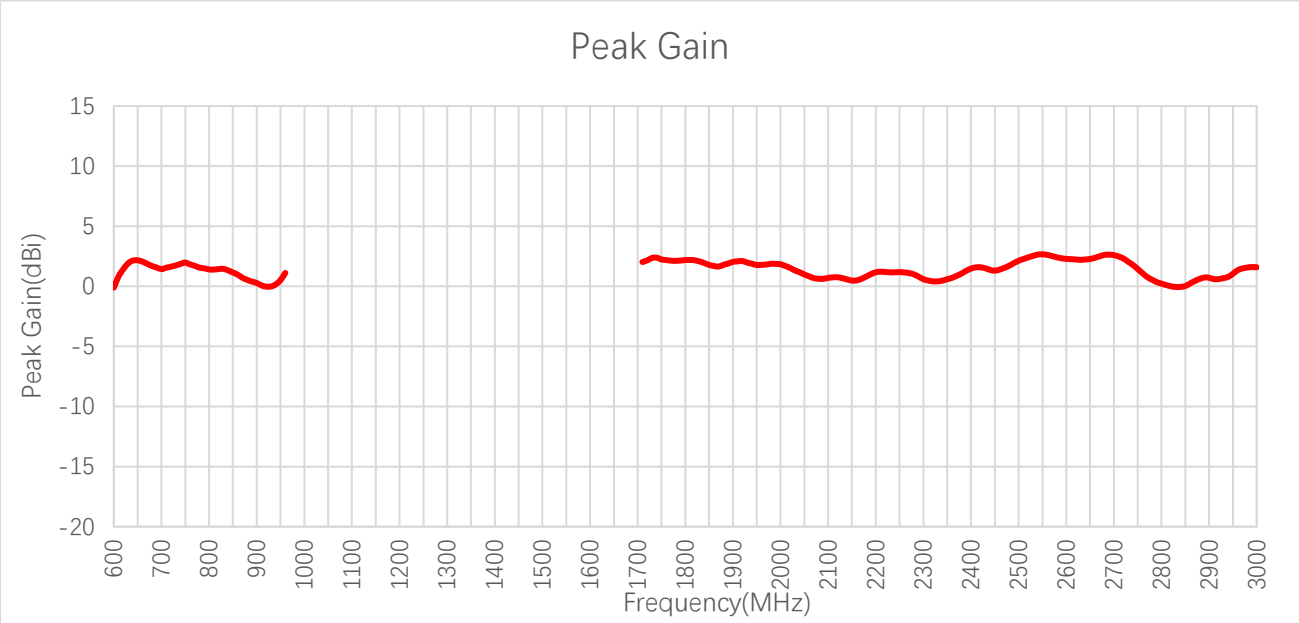
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Average Gain (dB)	-5.2	-3.6	-1.5	-2.5	-4.2	-5.0	-	-2.7	-2.7	-3.0
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Average Gain (dB)	-2.7	-3.9	-3.6	-3.1	-3.1	-2.8	-	-	-	-

3.2.3. Peak Gain



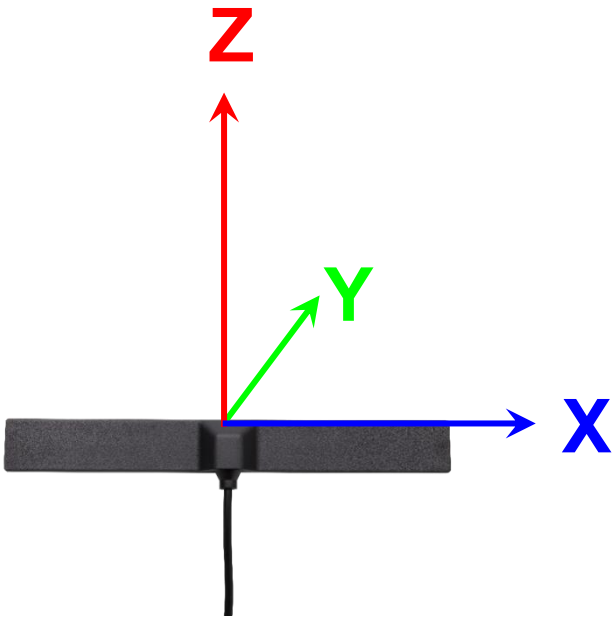
Peak Gain (dBi)

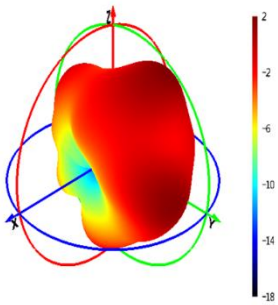
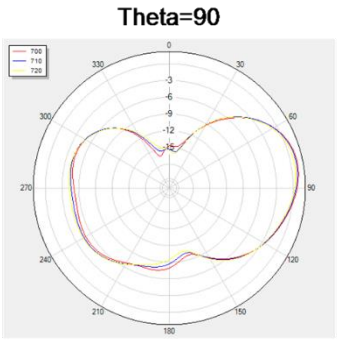
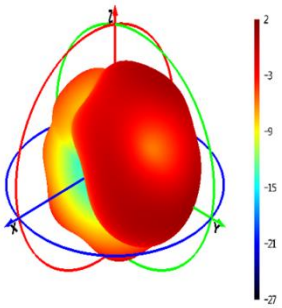
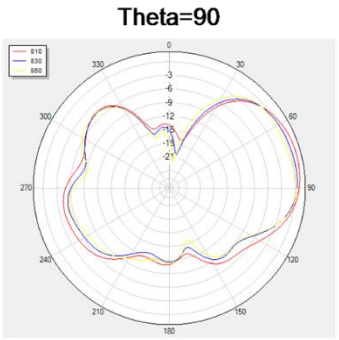
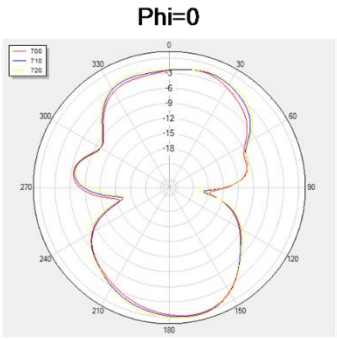
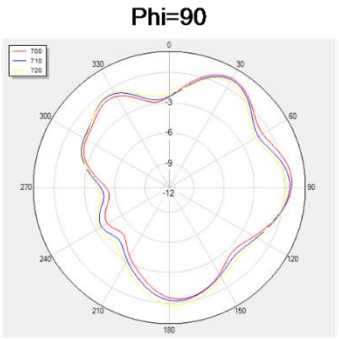
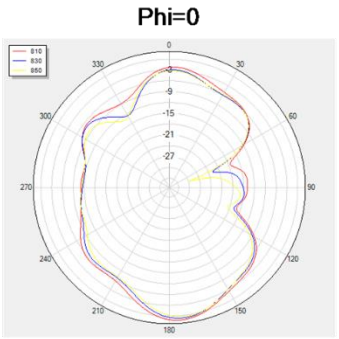
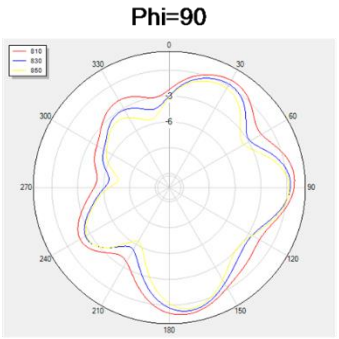
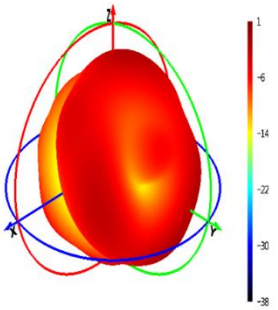
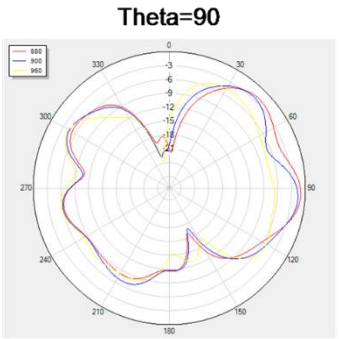
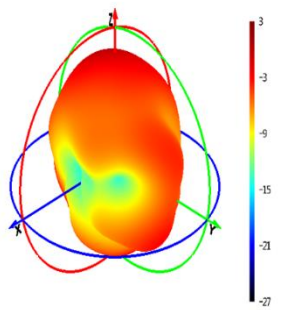
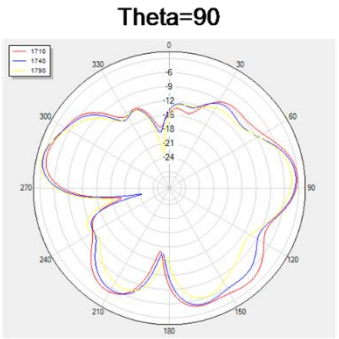
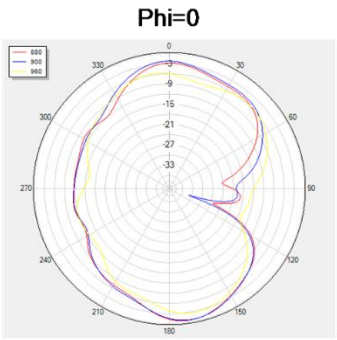
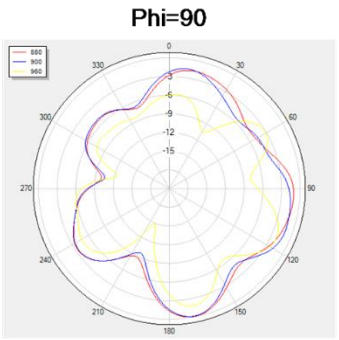
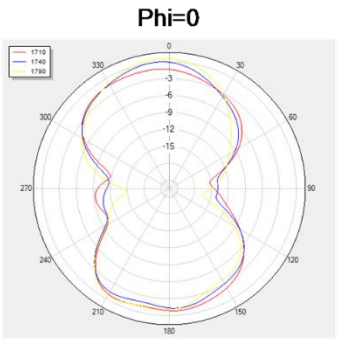
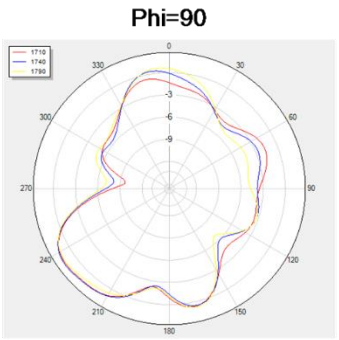
Frequency (MHz)	**600**	**630**	**710**	**830**	**900**	**960**	**1440**	**1710**	**1740**	**1880**
Peak Gain (dBi)	-0.1	1.9	1.5	1.5	0.3	1.1	-	2.0	2.4	1.8
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Peak Gain (dBi)	1.8	0.6	0.6	1.3	2.3	2.6	-	-	-	-

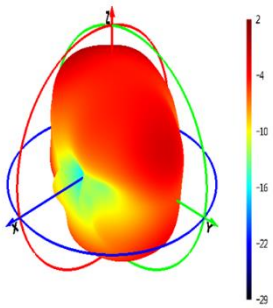
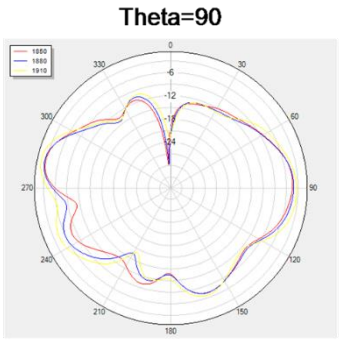
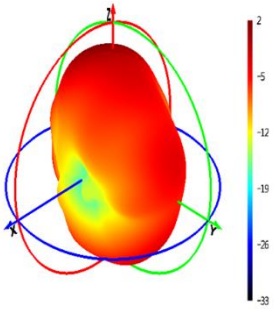
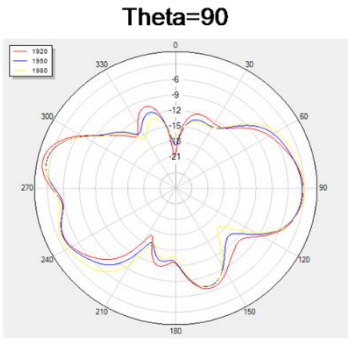
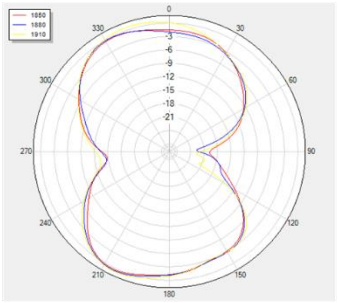
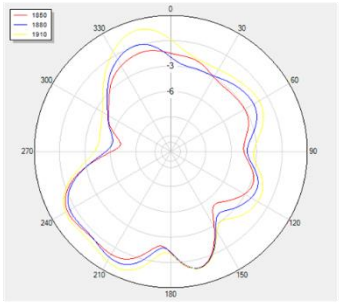
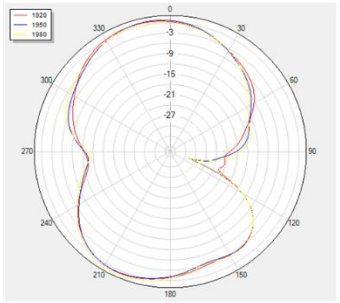
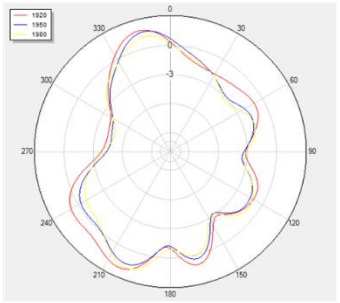
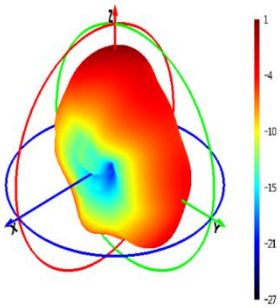
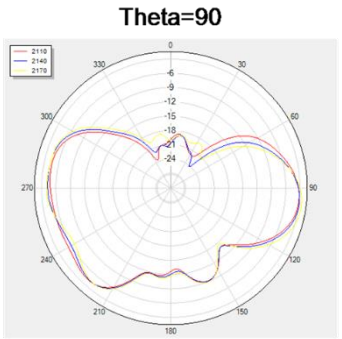
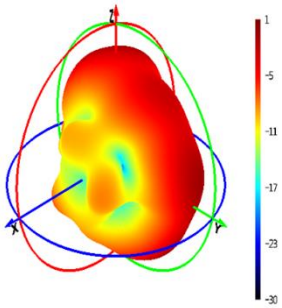
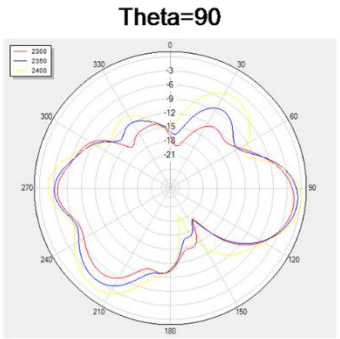
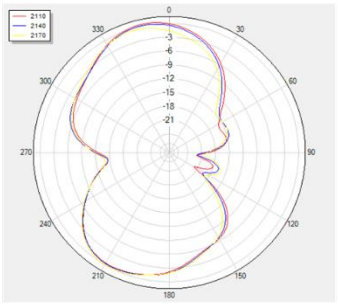
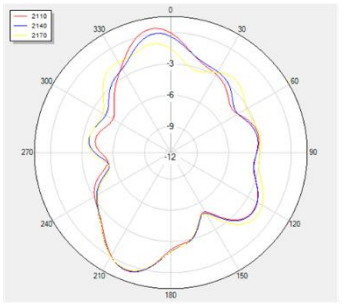
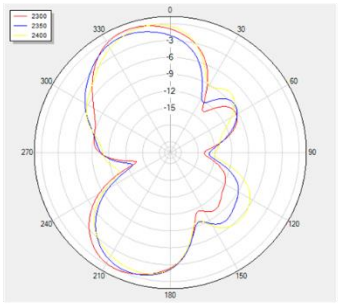
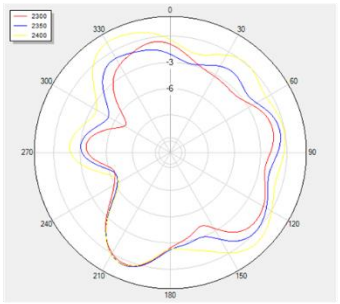
3.2.4. 3D & 2D Radiation Pattern

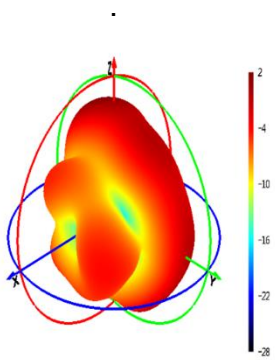
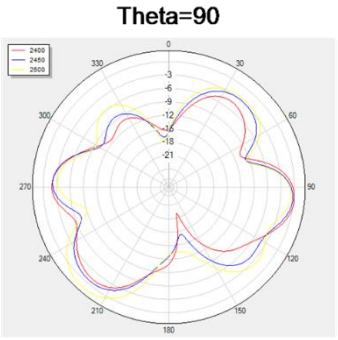
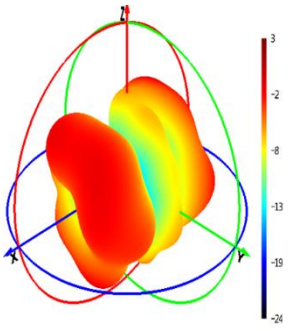
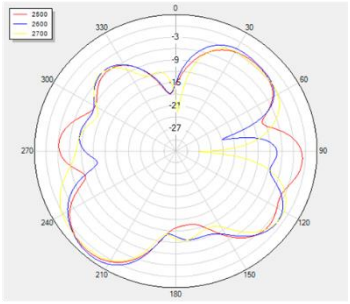
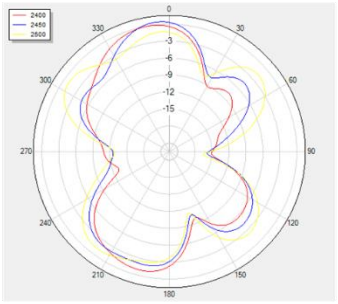
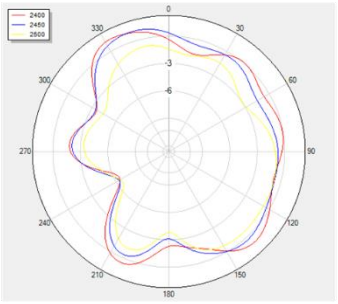
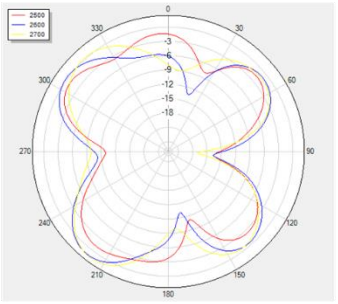
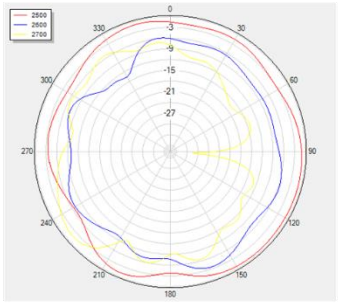
3.2.4.1. Test Condition: Free Space

Test Chamber: GL-S-1



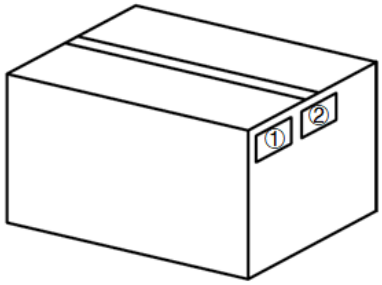
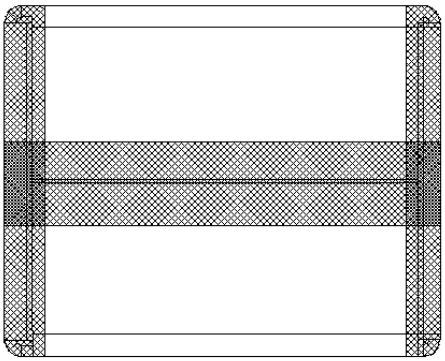
710 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

1880 MHz		1950 MHz	
			
			
2140 MHz		2350 MHz	
			
			

2450 MHz		2600 MHz	
	Theta=90 		Theta=90 
Phi=0 	Phi=90 	Phi=0 	Phi=90 

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		1 antenna product in a PE bag.
2		20 antenna products in a big PE bag.
3		(8 PE Bags / Carton Box) (160 Antennas / Carton Box) Carton Size: L × W × H = 405 × 293 × 185 mm

4		**Position for Attaching Labels** ① Carton Label ② Quality Label
5		**Sealing Cartons** H-shaped sealing cartons

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

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Revision History

Version	Date	Author	Note
-	2020-06-04	Kenny Yin	Creation of the document
1.0	2020-06-04	Kenny Yin	First official release
1.1	2020-12-11	Kenny Yin	Updated the antenna image (Chapter 2).
1.2	2021-01-27	Kenny Yin	Added IP rating description and installation method.
1.3	2021-07-25	Kenny Yin	1. Updated working temperature (Chapter 3). 2. Added detailed passive electrical specifications (Chapter 3).
1.4	2021-12-01	Kenny Yin	Updated the product description (Chapter 1).
2.0	2023-05-11	Mikael Zhong/ Lucky Feng/ David Liu/ Aria Chu	Updated all data and datasheet template.
2.1	2024-01-15	Lucky Feng	1. Updated the antenna size and added storage temperature (Chapter 1.2). 2. Updated the drawing (Chapter 2).
2.2	2025-04-09	Rainey Liao	Updated the antenna image (Cover page).
2.3	2026-05-18	Lucky Feng/ Rainey Liao	1. Added the overview. 2. Updated the drawing (Chapter 2).



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